

Dynamical Oceanography

Problem Set 2

Due Monday March 1, 2010

1. When deriving the solution to the Stommel problem, we made the choice $\epsilon = \epsilon_S$, where ϵ was the stretching parameter and ϵ_S was the nondimensional frictional parameter. Explore the consequences of choices of the form $\epsilon = \epsilon_S^\gamma$ where, for example, $\gamma = 2$ or $\gamma = 1/2$. Can obtain a solution that satisfies the boundary conditions and other requirements? Is $\gamma = 1$ the only value that works, and is this solution unique?
2. Obtain a boundary-layer solution of the Munk problem [(14.42) of AOFD] but with ‘free-slip’ boundary conditions instead of no-slip conditions. You may use a simple single gyre wind stress of your choice. If the algebra defeats you, explain carefully what you are trying to do.
3. Consider the steady nonlinear Stommel model in the form:

$$J(\psi, \zeta) + \beta \frac{\partial \psi}{\partial x} = \nabla \times \tau - r\zeta \quad (1)$$

Show that the nonlinear term will be small in the interior of the domain if $\beta L^2/U \gg 1$ where L is the scale of the large-scale motion and U a representative velocity. Show (e.g., by using scale analysis) that for sufficiently strong wind, the nonlinear term must become more important. Derive a criterion (in terms of τ and the other parameters in the equation) for this. *Either* do question 5 *or* describe qualitatively how you might obtain nonlinear corrections to this, and what the nature of the corrections might be. Briefly discuss the utility of this — how accurate the solutions might be, where they might fail, and so on.

4. Suppose that nonlinearity were expected to be of leading order importance in the ‘non-linear Stommel-Munk problem’ (e.g. in (14.48) of Vallis, AOFD). In a square domain of side L , show that this suggests that the non-dimensional version of the equation should be:

$$J(\hat{\psi}, \hat{\zeta}) + \gamma \frac{\partial \hat{\psi}}{\partial \hat{x}} = \epsilon'_S \nabla^2 \hat{\psi} + \text{curl}_z \hat{\tau}_T + \epsilon'_M \nabla^2 \hat{\zeta} \quad (2)$$

where $\gamma = \beta(L^3/|\tau|)^{1/2}$ and $\epsilon'_S = r(L/|\tau|)^{1/2}$. What is ϵ'_M ? Friction seems to become less important as τ increases. Is there a limit in which friction is nowhere important?

5. Optional: Using standard perturbation theory (e.g., by supposing that the solution is given by: $\psi = \psi_s + \psi'$ where ψ_s is the solution to the Stommel problem and ψ' is a small correction) explicitly obtain the nonlinear corrections to the Stommel solutions for a single gyre flow. Show that the effects of increasing nonlinearity are to move the center of the gyre poleward, and discuss the other properties of the solution. Where are the errors largest? [This problem is made much easier by using algebraic manipulation software, such as Maple.]
6. Optional: Obtain solutions of the Stommel model in a circular or triangular domain.